CHEM E 480 Lab 1 Reflection

1. Describe the components on the Arduino hardware. Which of these are inputs and which are outputs?

The main parts I used on the TCLab were the two temperature sensors, T1 and T2, and the two heaters, Q1 and Q2. The temperature sensors are the inputs because they send temperature measurements back to Python. The heaters are the outputs because I control them with the code. The Arduino board and USB connection allow the computer to send commands to the device and read the sensor data.

2. Describe the purpose/action of each line of code.

- Line 1: I set n = 300, which means I collect 300 one-second intervals, or 5 minutes of data.
- Line 2: I create tm with np.linspace(0, n, n+1), which gives me the time array in seconds from 0 to 300.
- Line 3: I create the lab object with tclab.TCLab(), which opens communication with the TCLab hardware.
- Line 4: I start the list T1 with the current measured temperature from sensor 1, in °C.
- Line 5: I set heater 1 to 50% power with lab.Q1(50), which is the step input.
- Line 6: I start a loop that runs once per second for n total seconds.
- Line 7: I pause the code for 1 second so the measurements are taken at one-second intervals.
- Line 8: I append the newest T1 value to the list so I store the temperature history over time.
- Line 9: I close the device with lab.close() so the connection and heater are shut down safely.
- Plot line 9: I create Figure 1 for the plot window.
- Line 10: I plot measured temperature versus time as red points and label it 'Measured'.

3. What is the solution to the ODE? How good of a model is it?

The solution I used from lab was T(t) = T_a + K_p Q (1 - e^(-t/τ_p)). I would say this is a decent first-order model because it captures the general heating trend and the approach to steady state, but it is still simplified, so it does not match the real data perfectly.

4. What is a step input? What effect did it have on the data?

A step input is a sudden change in the input from one constant value to another. In this lab, setting the heater to 50% or 100% was a step input. When I applied it, the temperature did not jump instantly. Instead, it rose smoothly over time toward a new steady-state value. When the heater stepped back down, the temperature decayed back toward ambient.

5. In the oscillations section, describe the temperature response and explain why it happened.

When the heater input oscillated, the temperature also oscillated, but with a smaller amplitude and a time lag. I saw that the temperature response was smoother than the heater signal and did not follow the input immediately. That happened because the TCLab has thermal inertia and a finite time constant, so the system behaves like a first-order process, filtering out fast input changes. As a result, the temperature never fully reached the maximum before the heater changed again.